

An analysis of Intrusion detection systems in IIoT

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Abstract— Enhancing manufacturing and industrial processes are achieved using smart sensors and actuators known as the industrial internet of things or IIoT. By implementing IIoT, machines transform into smart devices, communicating important information that enables driving business decisions faster and more accurately. Thus, IIoT enables the improvement of business processes that are agile and dependable. However, the widespread deployment of IoT networks across different industrial sectors has resulted in numerous security gaps, posing threats to the integrity of IIoT systems. An intrusion detection system (IDS) is a typical software that monitors and stimulates security explanations for computer networks. Deployment of the solutions aims to identify malicious activity and take measures that facilitate risk aversion. Due to IIoT's peculiarities, standard IDS-based solutions are complicated to implement. This includes limited resources, sensitive data, and heterogeneous architecture. Researchers are implementing Fog/Edge computing, Machine Learning (ML), and Deep Learning for effective and flexible IDS deployments in different IIoT scenarios. This research focuses on how IDS emerges in specific manufacturing settings. To do this, we present an in-depth analysis of IDS deployment strategies, detection strategies, evaluation approaches, and data sources.

Keywords— An intrusion detection system (IDS), Internet of Things (IoT), Industrial Internet of Things (IIoT), Cyber Security, Machine learning (ML), Deep Learning (DL).

I. INTRODUCTION

Industry 4.0 has emerged as a new business paradigm using IoT technologies in new and innovative ways. Cyber-Physical Systems (CPS) and other ICTs allow intelligent machines to automate processes to a certain extent. As such, this is the kernel of operational technology. However, hackers utilize IoT devices as hosts for their malware to carry out a variety of destructive cyber threats because IoT networks are set up in many separate businesses that aren't connected and because IoT devices don't have the necessary security built in.

However, OT has been attacked; it is responsible for managing IIoT processes and has been the target of several cyberattacks with potential physical-world consequences. The proliferation of this fresh perspective and the full realization of realization.0 are hampered and need to be improved concerns. Security experts and academics agree that an Intrusion Detection System (IDS) should be used to enforce security policies by identifying potential vulnerabilities appropriately. This would prevent privacy, service quality (integrity), and productivity issues in Industry 4.0 Information systems, such as those used in the Industrial In hybrid IT/OT settings, control systems (ICS) and mission-critical infrastructure are of paramount importance. The plan aims to analyze behavior for anomalies and prevent intrusions before they happen. The following factors make it challenging to develop an IDS suitable for the IIoT: The vast and varied data sets produced by IIoT make information difficult to locate. Most ML-based detection techniques are limited to edge devices in the IoT because they require a larger computational footprint for data processing and model learning. Finding modules related to data privacy is another example of a problem that necessitates highly specialized and complex algorithms. An adaptive IDS should be able to detect any cyber threats with as few fake alarms as possible before they cause destruction [2], and the home environment shouldn't interfere with the primary services. An effective IDS for IIoT environments by combining several different detection methods and strategies. It protects industry 4.0 from growing cyber threats and any new use of its vital resources. Therefore, it is essential to deliver an up-to-date universal evaluation of IDS-based cyber security for IIoT, detailing the issues and factors that researchers must consider when developing a system that meets the requirements [3]. Among the variables influencing this study are the following:

- Identifying security issues using IDS-based cyber security solutions and IIoT setups.
- Identifying potential issues with IDS-based

cyber security in the IIoT.

- Determining the IDS detection techniques frequently applied in an IIoT scenario.
- Determining and assessing proposed detection strategies based on the suitable performance indicators frequently utilized for intrusion detection in IIoT.

II. RELATED TASK

Regarding cyber security, IDS is a good way to protect networks and information systems. Academics are interested in the new technologies for detecting intrusions developed recently as part of the IoT strategy. Pundir et al. looked at the security problems, and attacks that can happen in the Internet of Things (IoT) and Wireless Sensor Network (WSN) based transmission environments. It reviewed how WSN and IoT use Protocols, models of potential danger, and architectural designs. Their paper also compared how well IDS approaches worked. The IIoT Cyber-Physical System (CPS) controls the physical world [4]. ML-based IDS for IIoT technologies like IoT, wireless communication networks, and cyber-material systems has been reviewed. Before considering ML-based networks and IoT intrusion detection, Da Costa et al. talked about security concerns in the IoT. In their study, they only looked at the detection strategy, the target protocol, the accuracy performance, and the used dataset, which is needed to determine the IDS's effectiveness more accurately[5]. Elrawy et al. examined the used IDSs for the IoT paradigm after discussing the usual IoT architecture, related threat types, and growing security vulnerabilities. Their investigation was conducted using the following criteria: placement, performance metrics, detecting method, detection methodology, and some attributes. The primary IDS criteria were clearly stated, and the proper IoT needs were highlighted. On the other hand, validation data should have been considered [6]. The data source, however, should have been considered. In their study of several chosen IDSs for IoT networks, [7]Santos et al. classified the IDS based solely on these characteristics: detection method, security thread, and placement approach. The range of IoT technologies and new assaults is not covered in their chosen works. Also, their study only used a small set of categorization attributes, which doesn't capture IoT IDS security's true nature. Zarpelao et al. evaluated IoT IDS systems based on recognition method, placement strategy, protection

threat, and validation strategy. The study didn't reveal the main IDS classification detection strategy or performance indicators. Hajiheidari et al. examined Internet of Things intrusion detection methods. According to the literature, their study focused on detecting and detecting nine types of target attacks and related performance measures.

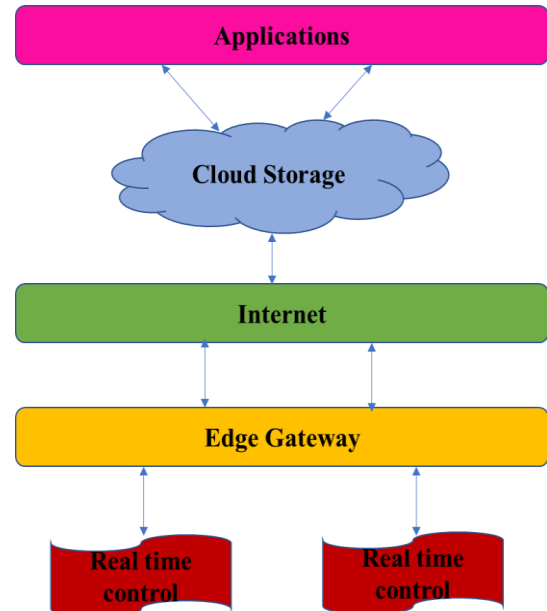


Fig. 1: Architecture of IIoT networks

The pros and cons of the subject matter are also brought out[9]. Mitchell and Chen examined the CPS-specific design principles and detection methods for IDS. They concentrated on two primary areas: audit material and detection methodology (data source). The CPS architecture's and current protocols' IDS functionality are fully explained. However, there needs to be a discussion of the detecting methods employed [10]. Most relevant publications have left out much important information about how IDS has changed, especially regarding IIoT. To safeguard industry 4.0 from developing cyber threats, our research aims to fill the void between the evolution of traditional IDSs and good IDSs for the IIoT by elucidating the necessary recognition strategies, trials, obligations, and essential attributes. Fig. 1 shows how IIoT networks are put together.

III. IIoT SECURITY PROBLEMS AND A SOLUTION BASED ON IDS

Industrial IoT architecture differs from typical. There are greater control system and security risks associated with IoT and CPS systems. A typical IIoT

architecture may be described as a hierarchical layer consisting of various schmoozing types of machinery and interaction protocols that link hardware gadgets to control software systems and end users. Communication technologies that could be stronger are a major worry for modern IIoT systems' security. IIoT networks are harder to protect from cyberattacks because they use proprietary layered protocols, different communication infrastructures, a high trust barrier, and few ways to upgrade. Because there aren't enough resources, it's almost impossible to make IIoT systems more secure while keeping them running. So, to protect IIoT environments from security threats, you need to build a good plan for monitoring security and finding attacks[11]. An IDS meets two security criteria for the IIoT: the ability to keep an eye on infrastructure and multiple cyber threat detection methods to address new threats. Building IDS detection algorithms and system architecture requires careful resource management. IIoT devices have limited storage, energy, and computing power and different data sources. IDS has been used in CPS, communication protocols, and public ICS infrastructure like water treatment, gas pipelines, and transportation in several IIoT studies. We provide an IDS taxonomy based on deployment and discovery. This will show how IIoT IDS systems differ from information system IDS systems.

IV. CLASSIFICATION OF IDS DEPLOYMENT SCHEME IN IIOT

This section discusses IIoT network IDS placement and interaction. We'll evaluate IDS performance and find issues next. IDS architecture can monitor and analyze Industry 4.0 and protect plant networks from cyberattacks and other proceedings that can be predicted by connecting CPS and IoT explanations using cloud and fog concepts. Data privacy and using IIoT resources for IDS computation make security monitoring IDS setup difficult. Low overhead protects confidentiality and can respond in real time to large-scale cyber threats involving the IIoT. Depending on whether the data from the monitored area were collected in one place or many places. An IDS could be at the same node where the data come from or at more than one node.

Centralized/Decentralized IDS: Indicates whether the IDS is installed on a separate node or a collection of IDS representative nodes, each of which makes its own decisions based on its observations of the local

environment. Centralized IDS, according to research by Aydogan et al. [13], can better detect attacks in real time because it gathers more data than individual IDS agent nodes. When it comes to IIoT resource issues, it is also superior. Most commonly, cloud and fog. By housing their accumulated data and costly processing computations, computing enables centralized IDS. However, decentralized IDS is a solution that can continue to function despite interruptions. In addition, heterogeneous IIoT's multiple detection models should aid in discovering widespread attacks without endangering users' privacy.

Dispersed Intrusion Detection Systems: Implemented on multiple IDS agent nodes, all of which collaborate to help make decisions about the IDS based on the data they collect. In edge computing, the distributed IDS helps with jobs that need many resources but only a small amount of data to look at. The sources showed an intrusion detection system (IDS) based on data that had multiple layers, a defense-in-depth strategy, and a wider range of attacks that could be detected. The authors suggested a way to find anomalies with more than one level and use various detection algorithms, one for each tier. Independent Identically Distribution (IID) and non-IID data were associated by Shu et al. to see how well distributed IDS worked [14].

Hybrid-based IDS: IDS input data are important in finding large-scale attacks. A hybrid IDS uses both host data (host-based IDS) and tangible information (state-based IDS) to quickly find out what is wrong, while a network-based IDS (NIDS) uses network traffic data to find out about attacks on the network. A command tampering attack or a fake data injection attack by an insider can be stopped by a hybrid-based IDS in an ICS. To find intrusions that monitoring data from the web and the host system might not catch. Zhou et al. made different data models to capture the wide-ranging information of Industrial Process Management Systems so that hybrid anomaly-based IDS (PCS) could be used [15].

CLASSIFICATION OF IDS DETECTION SCHEME IN IIOT

We go through IDS detection techniques in the IIoT in this part. We specifically talk about the detecting approaches and methodology. The IDS solutions for IIoT networks are categorized in Fig. 2. In Table.1, a

taxonomy of Methodologies for IDS detection in IIoT networks.

A. Methods of Detection:

Signature-established detection can effectively thwart malicious intrusions by keeping a database of previously identified attack signatures and using signature matching to predict when those attacks will occur—finding previously identified attacks in real-time cost-effectively and efficiently. It can't handle polymorphic or unknown attacks and needs human help, secures connections with enough bandwidth for remote updates and constant updates for new attack signatures [16].

User behavior analysis (UBA)-based anomaly detection: develop a detection model in which a user (a piece of hardware) is the focus of the analysis., software, or a person. Research is done on the network traffic, the devices used, and the operating system. A very effective and self-learning way to find attacks you don't know about. False positives are more common and are more demanding on the computer. [17].

Specification-based detection: Analyze a protocol or system to define a model of acceptable behavior while keeping an eye on the performance of a feature called specs to find any deviations from these requirements. It is good at detecting attacks and can be trained without training. Even though it might need more computing power to look at how conditions behave, it can't find episodes that fit the requirements model. With a hybrid method of detection, you can get better accuracy, fewer false alarms, lower costs, and detection in real-time. As an example of a hybrid IDS outline, Otoum et al. [18] suggested using IIoT gateways to solve the problems mentioned earlier and reduce the amount of work

needed to find intrusions.

Analyze a protocol or system to establish a model of appropriate behavior, and then keep an eye on how specs are performing to spot any deviations from these specifications. It requires no training and is effective at recognizing attacks. It may demand more processing capacity to analyze requirements behavior, but it cannot identify assaults that adhere to the specifications model. Presented an IDS subsystem that would first look for certain signatures in traffic packets to detect known attacks, then use anomaly-based detection to find unidentified attacks. To enable self-learning IDS for upcoming prognostications, extracted signatures are retained. Feng and Chana put out a similar approach [19], in which a standard signature tune catalog of the ICS network's field devices' known communication patterns was created and later used for time-series anomaly identification.

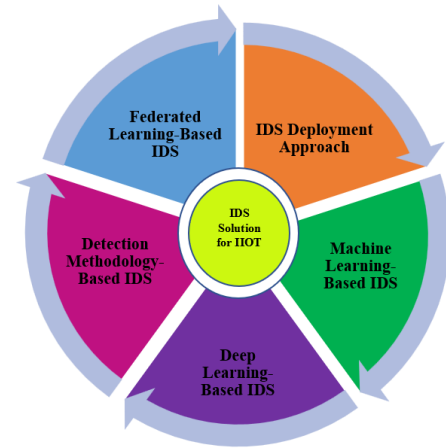


Fig. 2: IDS resolutions classification for IIoT networks.

Table I: Summary of IDS approach for IIoT

Ref. No	Techniques used	Dataset used	Performance	Comment
[20]	using an optimization and clustering model for multiple features of the data.	Both the NSL-KDD and KDDCup99	Accuracy - 97.8%,	The model needs to be tested with recent traffic data from industrial networks to see how well it works.
[21]	It uses a biologically inspired learning model to extract useful features and improves clustering based on	KDDCup99	Accuracy - 92.2%,	There need to be efforts made to reduce the time complexity of the biologically inspired learning model. As a result, the model's detection time may be longer than other IDS models.

	IDS systems			
[22]	DL fusion learning with different single CNNs and correlations is used for feature extraction.	NSL-KDD	Accuracy - 82.9%,	The model's effectiveness with IIoT network traffic datasets needs to be evaluated.
[23]	This method employs a federated deep learning scheme based on CNN and gated recurrent units.	real industrial CPS	98.64%	Applicable only for the same domain industrial CPSs
[24]	Deep random neural network	UNSW-NB15	Accuracy- 99.5%	The model is assessed for data from IoT networks rather than for traffic from industrial IoT networks.
[25]	Using a DL, there is adaptive trust boundary protection for IIoT networks.	Real-time	Accuracy- 96.6%	Cyberattack prevention for IIoT networks is hard because there is a big trust gap, and there are few ways to update.

B. Federated learning-based IDS

Federated learning (FL) is a new type of cooperative learning and distributed computing used in applications based on machine learning. The goal is for models to be sent from a central server. Instead of transmitting the client's data to the attendant, it could be sent to specific consumers for training with their information that matches. This would protect the client's privacy and let people learn from each other. FL-based IDS is a new idea for a consistent self-assurance model for IIoT's security needs and problems. FL allows privacy protection, security knowledge sharing, decentralized IDS decision-making as opposed to unrelated IIoT architecture, and effective anomaly-based IDS against sparse and scattered IIoT data [26]. Figure 3 shows a model of an Industrial IoT network next to how an FL-based IDS system is set up.

Clients: At the physical layer, where IoT devices always transmit data, you'll find the IIoT end nodes. To develop a regional detection model, each client would examine its data in isolation. The server would change the global model parameters and only share gradients made in the last round.

A Server for Aggregating: Client keeps informed are combined at the Point to make a single, international model which protects privacy. This model is then sent to the cloud attendant for storage and possibly Increasingly complex computation based on client-or publicly available data for analysis and maintenance.

Cloud server: captures the model and adjusts its global settings depending on information received between clients. In a distributed SDN context, where several SDN controllers train a single IDS, Shu et al. developed a collaborating intrusion detection system (CIDS) for vehicular ad hoc networks (VANETs).

Shu et al. [14, 27] made a to build a collaborative intrusion detection system (CIDS) for use in Vehicular Ad Hoc Networks (VANETs), the authors of this research suggest a distributed SDN environment in which many SDN controllers work together to establish a single IDS. Having a single IDS paradigm for the whole network allows them to avoid having their subnetworks exchange traffic directly with one another. Security for FL depends on cloud and edge computing, according to recent research by Fan et al. Private IoT traffic data is processed by the edge platform. In contrast, Transfer Learning (TF) from models trained on public data is handled in the cloud to protect privacy. In their proposal for an FL-based IDS, Nguyen et al. [28] suggested using individual detection models for various classes of devices. The Internet of Things Security Service provides an initial global model of all connected devices initial global model of all connected devices is provided by the Internet of Things Security Service to the security gateways. The security portals then utilize the traced data from the unstructured crowd-sourced traffic to train local models for each kind of device, therefore avoiding the costly server connection. Any updates to the models are sent to the server so they may be included. While FL is more discreet than

centralized learning, it does not outperform it in terms of precision. Malware detection in Android was addressed by the authors' FL framework, which was made public in [14], [27]. This was done to get around FL server issues. Byzantine protection measures were utilized to thwart the proposed attacks and identify flawed models. According to experimental results, the suggested defensive mechanism is superior to previous techniques published in the literature. Even though the fact that FL-based IDS ensures a very high level of confidentiality for your data, the shift to a new paradigm has introduced new IT and security worries. A few examples include the vulnerability of FL audit gateways to assault, the high cost of communication in large-scale deployments, and the problem of data and resource inequality.

C Detection strategies for IDS using Machine Learning

IDS algorithms that are based on machine learning have been used to protect IIoT networks and physical objects. These algorithms are often based on tasks and keep data distribution consistent. It makes traffic more predictable by addressing IDS detection efficiency in terms of how well it finds things and how well the system works in terms of how long it takes to decide. Derhab et al. [29]. Example blends Unintentional Subspace Learning (RSL) and KNN (RSL-KNN) to make good KNN classifiers that use different random subsections of selected features to increase detection accuracy. Khan et al. [30]. KNN, based on machine learning (ML), was used to find outliers as part of a multi-level detection method. In another paper, Reddy et al. (31) discussed an experimental evaluation of ML-based intrusion detection systems that used different ML algorithms to classify attacks and find outliers. Then, they suggested using XGBoost to make a good framework for group learning. The results showed that XGBoost is a better way to classify attacks than ML approaches for intrusion detection. KNN, bagging, and RF has been shown to offer ML-based IDS against Man in the Middle (MITM) and Distributed Denial of Service (DoS) threats in experiments.

D. IDS-based Deep learning

Deep learning is a subset of machine learning. It's a complex machine-learning process that can learn representations in layers. Although ML-based IDS is successful, DL-based IDS may be better suitable for IIoT security. When more individuals make use of

the system, it thrives. Intrusions/cyberattacks, difficulty, complexity, and distributivity may all be handled by various DL approaches. It was suggested by Fan et al. [32] that Deep Neural Networks (DNN) be used in a combined DL-based discovery method (IoT Defender) since DNNs are good at learning nonlinear data characteristics and transferring information. Other ML techniques, with or without TF and FL, couldn't compare to their results. Hassan et al. [25] partially implemented a DL-based attack detector. This was done to accommodate the ever-evolving attack models at the IIoT network's trust boundaries. To deal with the high volume of unknown attack packets in large-scale IoT systems, it was recommended to use deep Q-learning with reinforcement. To identify intrusions, the team at Muna et al. [33] developed a DL-based IDS that uses a supervised DNN for classification and an unsupervised learning Auto Encoder for tuning the system's parameters. This resulted in a more precise detection model. The LDAN demonstrated a low-power wireless sensor network using a light dynamic autoencoder (WSN). Safeguarding ICS using CNN anomaly detection. As Kravchik et al. proved that CNN could perform time series prediction tasks typically handled by recurrent neural networks, allowing for the detection of cyberattacks. To observe various network botnet assaults caused by hijacked IoT devices, the authors improved CNN learning parameters using bio-inspired Particle Swarm Optimization (PSO) [34]. CNN can identify intrusions by analyzing patterns in space-time. It can't determine chronological order or separate events. Long short-term memory (LSTM) and gated recurrent unit (GRU) recurrent neural networks (RNN) containing "memory cells" have recently been developed to detect and explain abnormalities in time series. In [19], the authors developed a layered LSTM model for learning high-dimensional temporal features, whereby outputs from one layer of the LSTM are fed into the next layer. The authors presented a GRU that can detect anomalies by analyzing packets of network data. Price was right, and not much was needed in the way of training data. To build reliable classifier models for intrusion detection in the face of malicious false training data intended to conceal illicit activity, researchers have turned to Deep Adversarial Learning using Generative Adversarial Networks (GAN). Aloul et al. (35) employed adversarial auto-encoders (AAE) and generative adversarial networks to build a model for intrusion detection at the IoT's periphery (GAN). Taheri et al. [27] showed two approaches (GAN and

FedGAN) that employ GANs to learn negative samples locally on the client to solve Adversarial FL issues.

Medium and Krum were used in conjunction with a GAN to manage the proposed attack scenarios and locate malicious models on the system. This was done to protect against attacks. Experiments have demonstrated that an attack based on a FedGAN is more reliable than an attack based on a GAN.

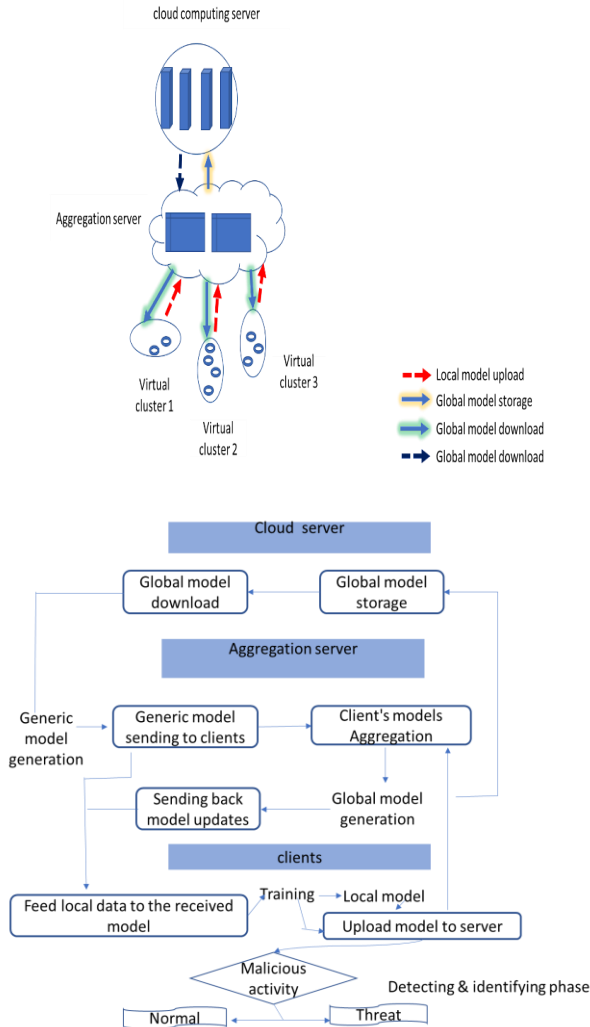


Fig. 3: The IIoT network model with an FL-based IDS system organizational chart.

Despite the success of using machine learning and deep learning-based detection approaches, there are still open questions about IDS-based security in the IIoT. Recognized. Minor changes to the data may throw off both ML and DL models, which turn can affect how well an attack is identified or categorized. It's hard to figure out where attacks are coming from and do more forensics because the decision-making process needs to be clearer and can't be understood.

The amount of learning and data processing required is still more than what the IIoT can do now. Also, adversarial attacks on model learning make it impossible to find malicious activity and give ML-based and DL-based IDS much trouble.

V. ASSESSING IDS EFFICIENCY

This part discusses IDS validation methods, how to evaluate performance, and where to get data. IDS appraisals are usually done with a validation method that can be theoretical, imaginary, pragmatic, or simulated, according to [8]. A validation strategy is a way to find out if the proposed IDS scheme is good for the job it was made for and follows all the rules. On the other hand, the evaluation checks to see if the IDS detection method meets the study's goals and keeps good performance metrics. All the papers were judged using simulations and tests in the real world. In the first case, real IIoT data is used to evaluate; in the second, real IIoT network traces are used. You can use these performance indicators to figure out how effective and efficient IDS is:

Accuracy: Explains how attacks are found correctly, and there are no false alarms. Many metrics are included, like accuracy, recall, true positive rate, false positive rate, true negative rate, and so on.

Complexity: This describes how much time and resources (memory, energy bandwidth) IDS activities like data processing and model learning take and how well they can be implemented and used in devices with limited resources. Most IDS ideas for IIoT need to take complexity metrics into account. Since there are no clear ways to measure, it is impossible to know how well and quickly the suggested solutions work.

Completeness: Describes how an IDS can effectively detect known and unknown threats equally and in reliability. This is most evident in the IIoT context when considering IDS's compatibility with large-scale organizations and its ability to handle data from various sources.

Maintaining IDS detection capabilities even as the number of IIoT behaviors increases is called scalability. IDSs can remember and use the profiles of events they have previously found to improve their detection abilities in the future.

All of the articles' performance statistics were added across various tests. A standard set of validation criteria is available by default to evaluate the viability of proposed solutions. Large datasets, both online and offline, are required to assess the efficacy of AI-based IDS. Academics have offered real industrial datasets like ICS, InSDN2020, SWaT, and TON IoT datasets due to the scarcity of simulated IIoT data due to privacy concerns. However, investigators still use classic IDS datasets such as NSL-KDD, UNSW-NB15, and CICIDS2017 to estimate the efficacy of their IDS-based protection concepts in IIoT. In most cases, this is done as a collection demonstrating the characteristics of various data types, the complexity of multiple cyberattacks, or the efficacy and efficiency of detection techniques in IIoT systems with constrained resources.

VI. IIoT IDS-BASED SECURITY: CHALLENGES AND OPEN ISSUES

IDS-protected in the IIoT faces a significant problem in developing an affordable and effective detection rate and conclusion latency detection methodology. We list related IDS problems categorized by their key traits for further study consideration:

1. Data source: Industrial data are dissimilar and large-scale by nature; they are typically derived from numerous sources using various networking technologies and communication protocols. IDS is difficult to handle heterogeneity and non-identically distributed data during data analysis and processing (N-IDD). Some industrial applications might need more data for big data processing or even anomaly-based IDS, which would impair IDS performance. Additionally, unbalanced data affects ML/DL-based IDS. In IIoT applications, privacy protection is a top priority and major concern.

Finally, the proposed IDS performances need more credibility due to the absence of real IIoT data and testbeds.

2. Methodology for detection: IIoT behavior analysis is difficult, and it's important to distinguish transient fault behavior from anomalies associated with security threats. No private information should be made public throughout the detection process (Privacy

protection). When the variety of behaviors increases, anomaly detection systems typically suffer from rising false alarm rates or falling sensitivity.

System Deployment: The IIoT's resource limitations limit the resources available for IDS solutions. The use of IDS might reveal new security holes.

Performance: Insecure IIoT communication protocols make it harder to foresee cyber threats and raise the likelihood of false alarms. Cost-effective IDS significantly impacts accuracy and real-time detection.

VII. FINAL THOUGHTS AND FUTURE RESEARCH

It is a workable and good idea to replace outdated models in contemporary IIoT schemes, as IDS systems are naturally flexible. In this paper, we reviewed the implementation of IDS for IIoT as part of rolling out cyber security. Later we discussed the limitations of IIoT and the issues linked to IDS detection methods, followed by a discussion on how the challenges, if resolved, would progress the IDS-based self-assurance system in IIoT networks' efficacy and efficiency.

As for the future research directions, the following could be pursued further:

- making IDS work across terminals, networks, and service platforms to drive a decentralized IDS design.
- Address the resource scarcity issues in IIoT as the deployment of distributed IDS architecture is expected to go smoothly.
- Assessing the findings of correlations from distributed IDS agent nod management as part of future studies is expected to be critical for the following reasons:
- Behavior analysis-based intrusion detection systems are an important research topic for various applications for the IIoT that provide self-adaptability and high predictability of intrusions.
- Expected new dangers in IIoT in the form of threats to data privacy even though ML and DL can improve IDS-based security protection; these issues could serve as a roadmap for research in the future.
- Enhancement of IIoT protection can be achieved by adding more primary mitigation measures

and IDS.

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